# SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

## Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

### Building Information

- Weather File: USA_NY_Buffalo-Greater.Buffalo.Intl.AP.725280_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 $m^2$

### Renovation Details by Scenario

| Scenario | Renovation Type | Renovation Details |
|---|---|---|
| Baseline | baseline | Baseline (no envelope renovation) |
| Scenario 1 | wall + roof + window + door | **Wall upgrade:**<br>• Exterior wall insulation: Fiberglass Batts, target R-20.4<br>**Roof upgrade:**<br>• Roof insulation: Blown Mineral Wool, target R-34.5<br>**Window upgrade:** |

| | | |
|---|---|---|
| | | • 1-pane replacement (skipped)<br>• Weatherstrip application (skipped: no OperableWindow)<br>• Glazing film application: safety film<br>• Caulking application: acrylic<br>• Window frame replacement: wood window frame<br>**Door upgrade:**<br>• Door replacement: wooden door (1 of 3 door(s) skipped: door type incompatibility)<br>• Bottom seal: automatic door bottom<br>• Top/side seal: jamb weatherstrip |
| Scenario 2 | wall + roof + window + door | **Wall upgrade:**<br>• Exterior wall insulation: Extruded Polystyrene (XPS) Foam Board, target R-20.4<br>**Roof upgrade:**<br>• Roof insulation: Expanded Polystyrene (EPS) Foam Board, target R-34.5<br>**Window upgrade:**<br>• 2-pane replacement (skipped)<br>• Weatherstrip application (skipped: no OperableWindow)<br>• Glazing film application: anti-graffiti film<br>• Caulking application: polyurethane<br>• Window frame replacement: wood-aluminium window frame<br>**Door upgrade:**<br>• Door replacement: polystyrene core steel door (1 of 3 door(s) skipped: door type incompatibility)<br>• Bottom seal: brush weatherstrip<br>• Top/side seal: silicone adhesive smoke gasket |
| Scenario 3 | wall + roof + window + door | **Wall upgrade:**<br>• Exterior wall insulation: Polyiso Insulation Foam Board, target R-20.4 |

**Roof upgrade:**

- Roof insulation: Polyiso Insulation Foam Board, target R-34.5

**Window upgrade:**

- 3-pane replacement (skipped)
- Weatherstrip application (skipped: no OperableWindow)
- Glazing film application: low-e film
- Caulking application: polyurethane
- Window frame replacement: wood-aluminium window frame

**Door upgrade:**

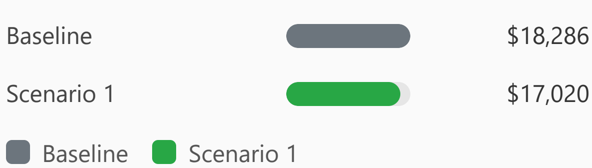
- Door replacement: garage door (skipped due to door type incompatibility)
- Bottom seal: automatic door bottom
- Top/side seal: silicone adhesive smoke gasket

## Annual Energy Analysis

| Scenario | Metric | Baseline | Renovation | Delta | Savings % |
|---|---|---|---|---|---|
| Scenario 1 | Total Site Energy (GJ) | 440.31 | 385.63 | **54.68** | **12.42%** |
| Scenario 2 | Total Site Energy (GJ) | 440.31 | 385.64 | **54.67** | **12.42%** |
| Scenario 3 | Total Site Energy (GJ) | 440.31 | 385.70 | **54.61** | **12.40%** |

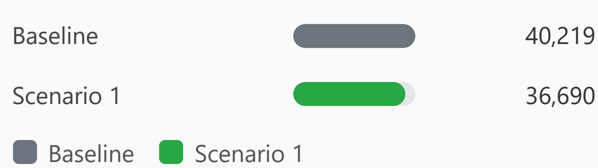
## Key Performance Metrics

### Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison




Max saving: **$-1,265 (-6.92%)**

### Operational Carbon Saving (Scenario - Baseline) (kg $CO_2e$/yr) Comparison



Max saving: **-3,530 kg $CO_2$e (-8.78%)**

Min Retrofit Construction Cost

**$20,270**

Scenario 1

Min Retrofit Embodied Carbon

**27,958 kg $CO_2$e**

Embodied carbon intensity (ECI): 55 kg$CO_2$e/$m^2$

Scenario 3

Lowest Retrofit Cost Payback Period

**16 years**

Scenario 1

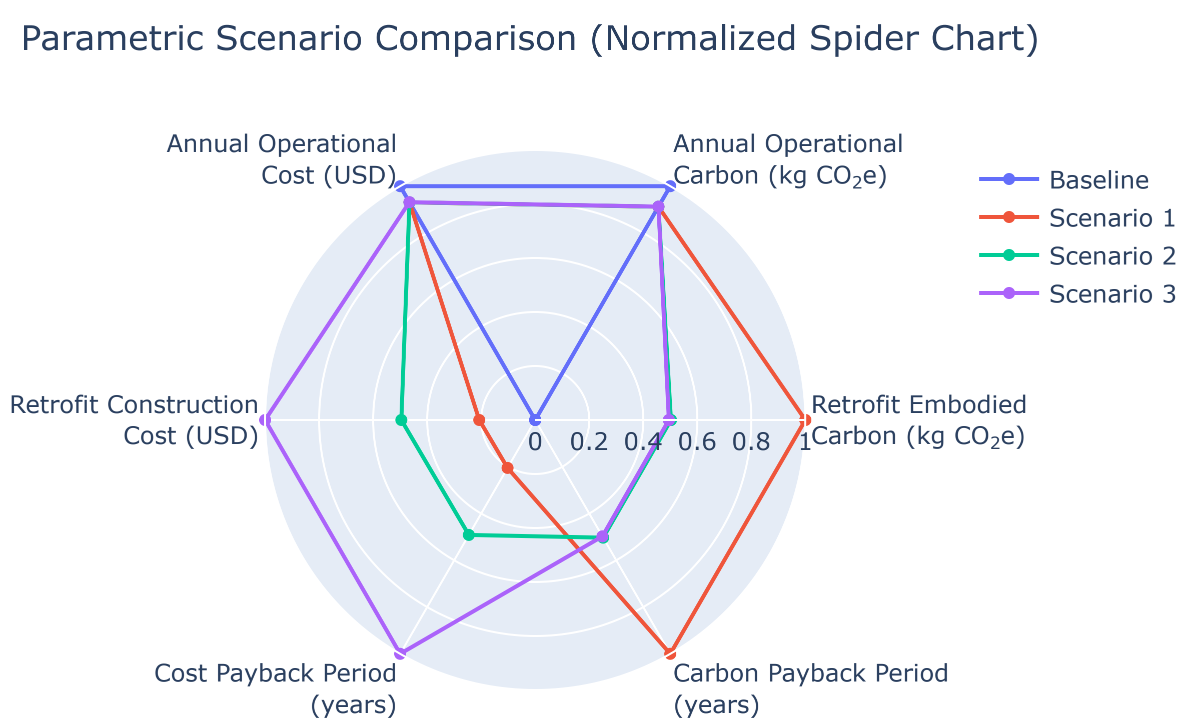
Lowest Retrofit Carbon Payback Period

**7.97 years**

Scenario 3

## Comparative Visualizations between Renovation Scenarios

### Spider Chart Visualization



## Payback Period of the Retrofit

### Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

| | |
|---|---|
| Scenario 1 | 16 yrs |
| Scenario 2 | 38 yrs |
| Scenario 3 | 78 yrs |

### Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg $CO_2e$/yr)).

| | |
|---|---|
| Scenario 1 | 16 yrs |
| Scenario 2 | 8.05 yrs |
| Scenario 3 | 7.97 yrs |

## Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

| Scenario | Component | Material name | Volume ($m^3$) | Area ($m^2$) | Thickness (m) | Length (m) | Thermal Conductivity (W/m·K) | Density (kg/$m^3$) | Product Lifetime (years) |
|---|---|---|---|---|---|---|---|---|---|
| Scenario 1 | Wall insulation | Fiberglass Batts | 17 | 282 | N/A | N/A | 0.03 | 30 | 60 |
| Scenario 1 | Roof insulation | Blown Mineral Wool | 144 | 599 | 0.24 | N/A | 0.03 | 90 | 30 |
| Scenario 1 | Window frame | wood window frame | N/A | 6.51 | N/A | N/A | N/A | N/A | 15 |
| Scenario 1 | Window caulking | acrylic | 0.0034 | N/A | N/A | N/A | N/A | N/A | 10 |
| Scenario 1 | Door | wooden door | N/A | 3.90 | N/A | N/A | 0.15 | 638 | 30 |
| Scenario 1 | Door bottom sealing | automatic door bottom | N/A | N/A | N/A | 3.66 | N/A | N/A | 15 |
| Scenario 1 | Door top/side sealing | jamb weatherstrip | N/A | N/A | N/A | 16 | N/A | N/A | 15 |
| Scenario 2 | Wall insulation | Extruded Polystyrene (XPS) | 17 | 282 | N/A | N/A | 0.03 | 35 | 75 |

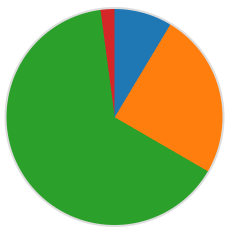
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Foam Board | | | | | | | |
| Scenario 2 | Roof insulation | Expanded Polystyrene (EPS) Foam Board | 144 | 599 | 0.24 | N/A | 0.03 | 20 | 60 |
| Scenario 2 | Window frame | wood-aluminium window frame | N/A | 6.51 | N/A | N/A | N/A | N/A | 15 |
| Scenario 2 | Window caulking | polyurethane | 0.0034 | N/A | N/A | N/A | N/A | N/A | 10 |
| Scenario 2 | Door | polystyrene core steel door | N/A | 3.90 | N/A | N/A | 0.10 | 472 | 30 |
| Scenario 2 | Door bottom sealing | brush weatherstrip | N/A | N/A | N/A | 3.66 | N/A | N/A | 15 |
| Scenario 2 | Door top/side sealing | silicone adhesive smoke gasket | N/A | N/A | N/A | 16 | N/A | N/A | 15 |
| Scenario 3 | Wall insulation | Polyiso Insulation Foam Board | 13 | 282 | N/A | N/A | 0.03 | 35 | 60 |
| Scenario 3 | Roof insulation | Polyiso Insulation Foam Board | 144 | 599 | 0.24 | N/A | 0.03 | 29 | 60 |
| Scenario 3 | Window frame | wood-aluminium window frame | N/A | 6.51 | N/A | N/A | N/A | N/A | 15 |
| Scenario 3 | Window caulking | polyurethane | 0.0034 | N/A | N/A | N/A | N/A | N/A | 10 |
| Scenario 3 | Door bottom sealing | automatic door bottom | N/A | N/A | N/A | 3.66 | N/A | N/A | 15 |
| Scenario 3 | Door top/side sealing | silicone adhesive smoke gasket | N/A | N/A | N/A | 16 | N/A | N/A | 15 |

# Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

## Scenario 1

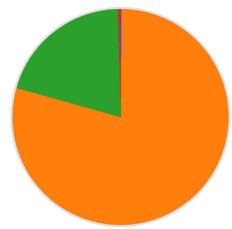
### Cost breakdown by retrofit measure



- Wall: $1,742 (8.6%)
- Roof: $5,021 (24.8%)
- Window: $13,082 (64.5%)
- Door: $426 (2.1%)

Total: $20,270

### Carbon breakdown by retrofit measure

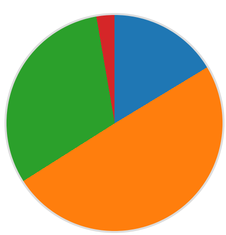


- Wall: 75 (0.1%)
- Roof: 44,758 (79.2%)
- Window: 11,437 (20.2%)
- Door: 241 (0.4%)

Total: 56,510 kg $CO_2e$

## Scenario 2

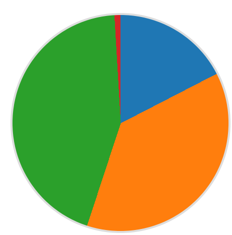
### Cost breakdown by retrofit measure



- Wall: $7,919 (16.3%)
- Roof: $24,099 (49.7%)
- Window: $15,154 (31.3%)
- Door: $1,318 (2.7%)

Total: $48,489

### Carbon breakdown by retrofit measure

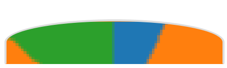


- Wall: 4,947 (17.4%)
- Roof: 10,684 (37.6%)
- Window: 12,496 (44.0%)
- Door: 265 (0.9%)

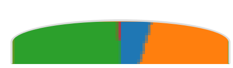
Total: 28,393 kg $CO_2e$

## Scenario 3

### Cost breakdown by retrofit measure



### Carbon breakdown by retrofit measure



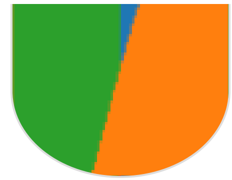
Wall: $6,959 (7.1%)

Roof: $77,651 (79.4%)

Window: $13,088 (13.4%)

Door: $132 (0.1%)

Total: $97,831



Wall: 1,244 (4.4%)

Roof: 13,882 (49.7%)

Window: 12,767 (45.7%)

Door: 65 (0.2%)

Total: 27,958 kg $CO_2e$

| Scenario | Retrofit Embodied Carbon (kg $CO_2e$) | Retrofit Construction Cost (USD) | Annual Operational Carbon (kg $CO_2e$) | Annual Operational Cost (USD) |
|---|---|---|---|---|
| Baseline | 0.00 | $0.00 | 40,219 | $18,286 |
| Scenario 1 | 56,510 | $20,270 | 36,690 | $17,020 |
| Scenario 2 | 28,393 | $48,489 | 36,694 | $17,024 |
| Scenario 3 | 27,958 | $97,831 | 36,709 | $17,034 |

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